Statistics and Trends

Student Mathematics Performance Analysis

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1. Introduction: Academic performance is important for understanding student progress. Mathematics grades provide useful insights into students' academic performance, particularly the relationship between first-period and final grades and the effect associated with previous class failures. These findings help identify differences in students' mathematics performance.

2. Data Description:

Table 1 : Dataset Overview and Selected Variables

Description	Details
Dataset name	student-mat.csv
G1	First-period grade
G2	Second-period grade
G3	Final mathematics grade
studytime	Weekly study-time category
failures	Number of previous class failures
Data types	Numerical and categorical

3. Statistical Analysis and Visualisation:

3.1 Histogram – Distribution of Final Mathematics Grades (G3)

As shown in Figure 1, the histogram illustrates the final mathematics grades of 395 students. Most students achieved grades between 8 and 16, with the highest frequency around 10–12. The mean grade is 10.42, the median is 11, and the standard deviation is 4.58. Fewer students achieved higher grades, while some received zero, showing variation in mathematics performance.

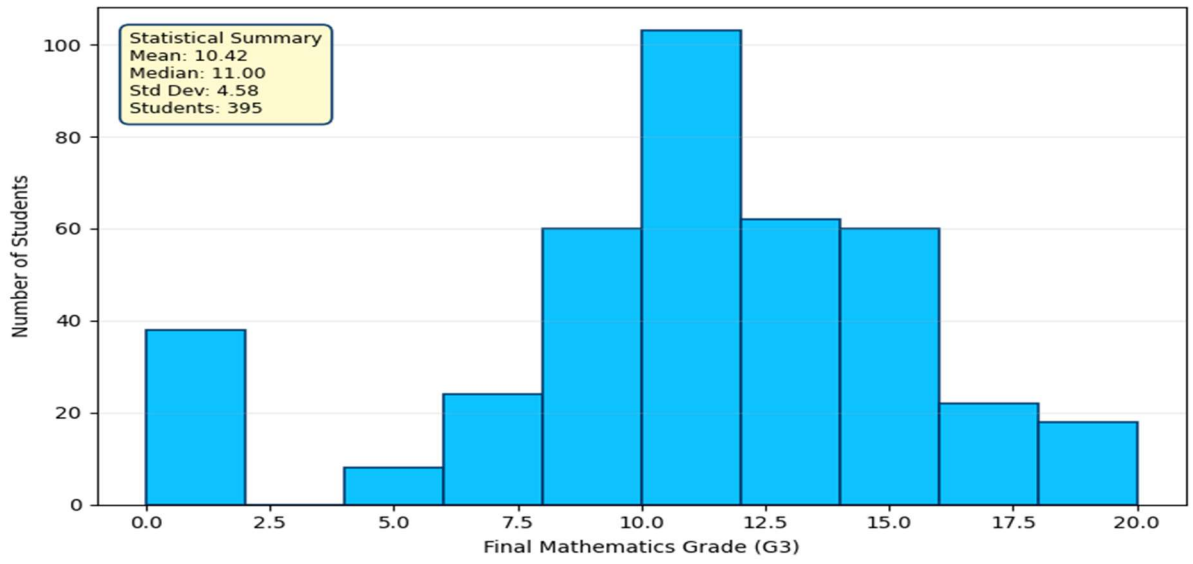


Figure 1: Distribution of Final Mathematics Grades (G3).

3.2 Scatter Plot – Relationship Between First-Period and Final Mathematics Grades

As shown in Figure 2, the scatter plot illustrates the relationship between first-period and final mathematics grades. The points generally move upwards from left to right. The correlation coefficient is 0.8015, indicating a strong positive relationship. Students with higher G1 grades tend to achieve higher G3 grades.

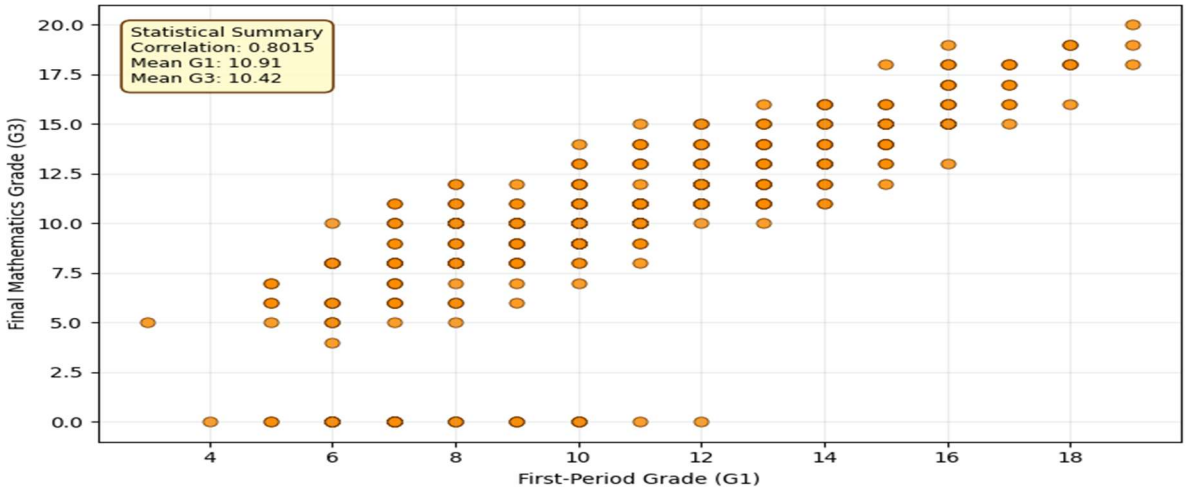


Figure 2: Relationship Between First-Period and Final Mathematics Grades (G1 vs G3)

3.3 Box Plot – Final Mathematics Grades by Previous Class Failures

As shown in Figure 3, the box plot compares students' final grades based on previous class failures. Students with no previous failures generally have a higher median grade. The median grade tends to decrease as the number of previous failures increases. Students with more previous failures generally show lower final grades.

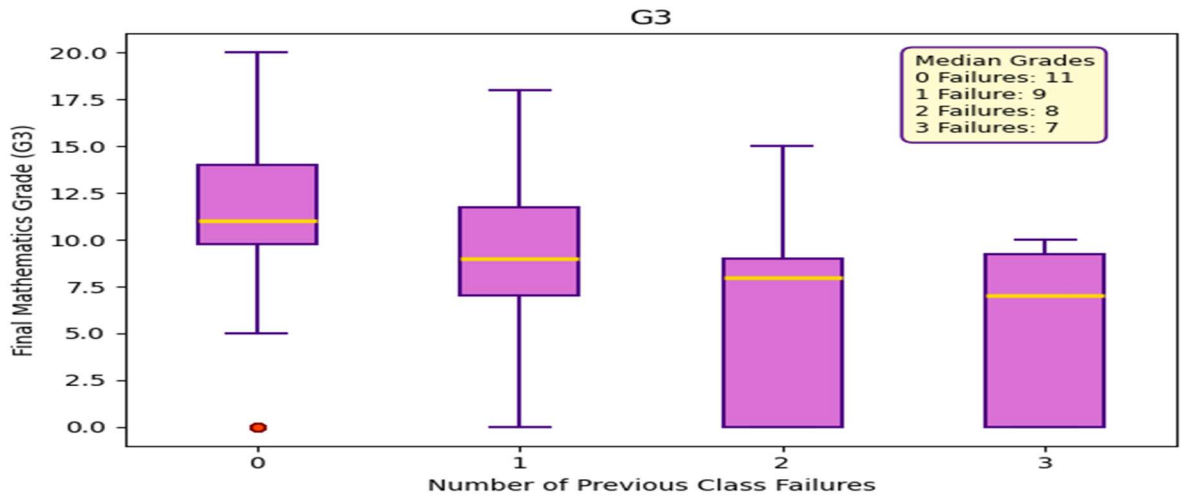


Figure 3: Final Mathematics Grades by Previous Class Failures (G3 vs failures)

4.Conclusion : The three plots provide useful insights into students' mathematics performance. They show grade distribution, the relationship between G1 and G3, and the impact associated with previous class failures.